# Exploring the effects of tree planting on simulated urban air pollution in Auckland


## Executive Summary

This project develops a simplified NetLogo model for Auckland Council's public education on tree planting and air pollution. Continuous emissions, wind transport and council-managed planting were added, with three experiments analysed in Python. Increasing planting intensity reduced pollution with diminishing absolute benefits. Wind changed pollution distribution and the additional benefit of planting. Targeted strategies performed better on average than random planting, but their relative performance varied with budget and between runs. The model illustrates how planting effort, environmental conditions and location influence outcomes. It supports public learning but requires local calibration and validation before informing practical planting decisions.


## 1. Background and Objectives

Auckland Council aims to increase average urban tree canopy cover to 30% by 2050[1][2]. Community participation and public education support this goal. Explaining how tree planting relates to air pollution is therefore relevant to the Council's environmental education work.

This project uses a simplified agent-based model (ABM) to explore how planting intensity, wind and planting location influence pollution. It aims to help the public understand the potential benefits and limitations of tree planting through three experiments:

1. How does planting intensity affect mean environmental pollution under fixed emission conditions?
2. How does wind affect pollution distribution and the reduction achieved by tree planting compared with no planting?
3. Under fixed wind conditions and equal tree budgets, how do planting strategies affect mean pollution and pollution hotspots?

The model reflects Auckland's management and educational context but uses abstract units for pollution, space and time. It is not intended to predict actual air quality. Visualised results provide an accessible explanation of model behaviour and show why both planting effort and location matter under the tested conditions.

## 2. Model and Experimental Methods

### 2.1. Model structure and development

This project adapts Wilensky's (2007) NetLogo model, Urban Suite - Pollution[3][4]. The model uses a grid of 71 × 63 cells, each storing a pollution value. Initially, all cells

have zero pollution, and two randomly selected cells act as fixed pollution sources. Trees remain where they are planted. Each cell can hold one tree, and source cells cannot be planted.

Three main changes improve the model's realism and educational value. First, sources add pollution each tick instead of resetting pollution to a fixed value. Second, directional wind transport moves a specified fraction of pollution to neighbouring downwind cells, alongside diffusion. Boundary wrapping is disabled, so pollution carried beyond the boundary leaves the model area. Third, resident-based planting is replaced by a council planting programme, including tree budgets and spatial planting strategies. Population agents and health, reproduction and movement processes are removed to focus on environmental pollution.

The diffusion coefficient is set to 0.8. Each tree removes up to 1 pollution unit from its own cell and 0.5 units from each neighbouring cell per tick. Pollution cannot fall below zero. These settings are retained from the original model (Felsen & Wilensky, 2007) and remain constant across all scenarios to compare the effects of planting intensity, wind transport and planting location. They use abstract model units and have not been calibrated against Auckland data.

**2.2 Experimental design**

All three experiments use two pollution sources to represent emissions from different locations while keeping the spatial setting simple. Their locations remain fixed during each run, and each source emits 0.1 pollution units per tick. This provides a constant pollution input for examining the effects of planting and wind transport. Grid dimensions, diffusion and tree removal capacity also remain constant across scenarios. The tested values are exploratory settings in abstract model units and have not been calibrated against Auckland observations. Table 1 summarises the experimental settings and recorded outcomes.

| Experiment ID | Variables tested | Settings and recorded outcomes |
|---|---|---|
| Experiment 1 | Planting intensity: 0, 0.5, 1, 2 or 5 expected new trees per tick | No wind; random planting; 100 ticks. Record mean pollution and tree count. |
| Experiment 2 | Wind transport fraction: 0, 0.1, 0.2 or 0.4; planting intensity: 0 or 1 new tree per tick | Eastward wind; random planting; 100 ticks. Record mean pollution and calculate paired reductions relative to no planting. |
| Experiment 3 | Tree budget: 5, 10 or 30 trees; planting strategy: random, source-targeted or high-pollution targeted | Eastward wind; transport fraction 0.2. Run without trees for 30 ticks, plant the full budget, then run for another 100 ticks. Record mean and maximum patch pollution at tick 130. |

**Table 1 Experimental settings and recorded outcomes**

Experiment 1 includes a no-planting baseline and a range of planting intensities to examine whether pollution reductions increase in proportion to planting intensity. The non-zero settings correspond to approximately 50, 100, 200 and 500 expected new trees over 100 ticks. Experiment 2 includes a no-wind baseline and progressively higher transport fractions to examine how wind transport affects pollution and the reduction associated with planting. The transport fraction represents the proportion of pollution moved in the selected direction per tick, rather than a measured wind speed. Eastward transport is held constant in Experiments 2 and 3 so that changes in wind strength or planting strategy can be compared without also changing direction[5].

Experiment 3 compares planting strategies under three limits on tree numbers, with equal budgets across strategies to isolate the effect of planting location. The initial 30-tick period allows a pollution pattern to develop, which guides the high-pollution strategy. All experiments are evaluated after 100 ticks of planting or post-planting simulation. These comparison periods do not establish steady-state conditions, and ticks do not represent actual years.

### 2.3 Simulation Replicates and Data Analysis

Each scenario uses 20 random seeds (1001 – 1020) to capture variation in source layouts and random planting. Paired comparisons in Experiments 2 and 3 match runs by seed, giving corresponding runs the same initial source layout.

Endpoint data are exported from NetLogo BehaviorSpace as CSV files and analysed in Python. Means, standard deviations, boxplots and paired differences describe average outcomes and variation between runs.

For Experiment 2, spatial snapshots are also exported at ticks 10, 30, 60 and 100, using the same initial source layout without trees, under no wind and a wind transport fraction of 0.2. These illustrate pollution spread in individual runs rather than average spatial patterns across repetitions.

## 3. Experimental Results and Sensitivity Analysis

### 3.1 Sensitivity to Tree-Planting Intensity

Under no-wind conditions and constant emissions, increasing planting intensity substantially reduced mean environmental pollution at tick 100 (Figure 1a). Without planting, mean pollution was 0.00447 units. At planting intensities of 0.5, 1, 2, and 5 expected new trees per tick, it fell to 0.00255, 0.00170, 0.000610, and 0.000132 units, respectively. These values represent reductions of approximately 43.0%, 62.0%, 86.4%, and 97.0% relative to the no-planting scenario.

The response was non-linear, with larger absolute pollution reductions per unit increase in planting intensity at lower intensities. For example, increasing intensity from 0 to 0.5 reduced mean pollution by approximately 0.00384 units per unit increase in intensity, compared with 0.000159 when intensity increased from 2 to 5.

Mean tree counts reached 49.65, 100, 200, and 500 across the four planting scenarios (Figure 1b). The diminishing benefit therefore did not result from tree numbers reaching a plateau.

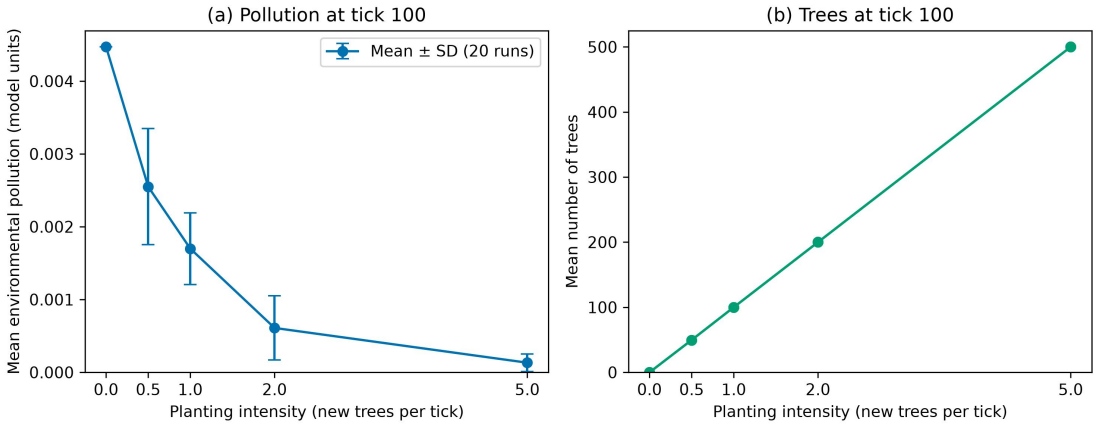



**Figure 1. Mean pollution (a) and tree count (b) at tick 100 across planting intensities.** Points show means from 20 runs; error bars in (a) indicate ±1 SD.

### 3.2 Sensitivity to Wind Transport Ratio

As the wind transport ratio increased from 0 to 0.4, mean pollution without planting decreased from 0.00447 to 0.00289 units, a reduction of approximately 35.4%. With one new tree planted per tick, mean pollution decreased from 0.00170 to 0.000644 units, a reduction of approximately 62.1% (Figure 2a). Pollution levels were therefore sensitive to wind strength in both scenarios, although the relative changes differed.

To assess the additional benefit of planting under each wind condition, pollution levels were compared between paired planting and no-planting runs (Figure 2b). At wind transport ratios of 0, 0.1, 0.2, and 0.4, the mean absolute reductions associated with planting were approximately 0.00277, 0.00282, 0.00262, and 0.00224 units. This response was non-monotonic: the reduction increased slightly at first and then declined. At a ratio of 0.4, the absolute benefit was approximately 19.0% smaller than under no-wind conditions. Planting produced lower pollution in all 80 simulation pairs, but the size of its benefit varied with wind conditions.

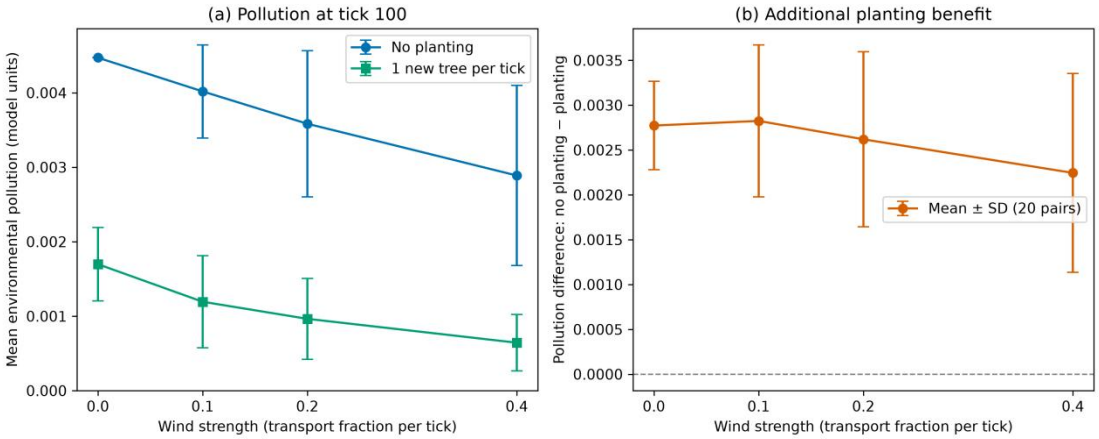



**Figure 2. Mean pollution (a) and paired pollution reductions from planting (b) at tick 100 across wind transport fractions.** Reductions are calculated as no planting minus planting. Points show means from 20 runs per scenario or 20 pairs; error bars indicate ±1 SD.

The spatial maps illustrate how wind changes pollution distribution (Figure 3). Without wind, pollution spreads around the sources; with eastward transport, it extends further downwind. However, wind can carry pollution beyond the model's open boundaries, which means lower pollution within the simulation area does not necessarily indicate removal by trees.

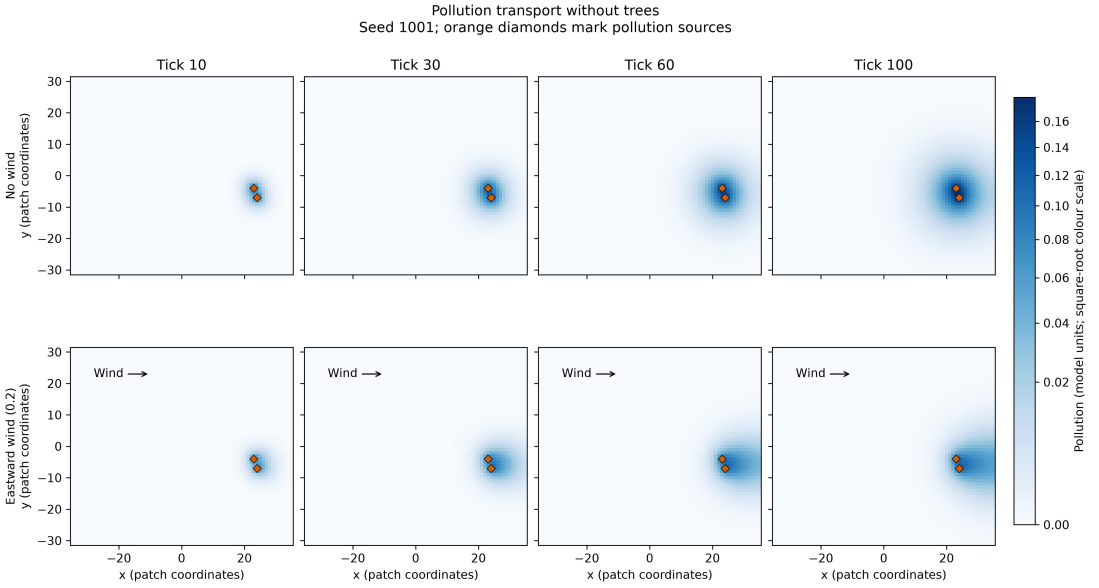



**Figure 3. Pollution distributions without trees at ticks 10, 30, 60 and 100, with no wind (top) and eastward transport of 0.2 per tick (bottom).** Both runs use seed 1001. Orange diamonds mark pollution sources; colour indicates pollution in model units.

### 3.3 Sensitivity to Tree Budget and Comparison of Planting Strategies

At each budget level, both targeted strategies produced lower mean environmental pollution and lower mean maximum patch pollution than random planting (Figure 4). Mean environmental pollution also decreased across all three strategies as the budget increased from 5 to 30 trees. Under random planting, it fell from 0.00391 to 0.00210 units, a reduction of approximately 46.4%.

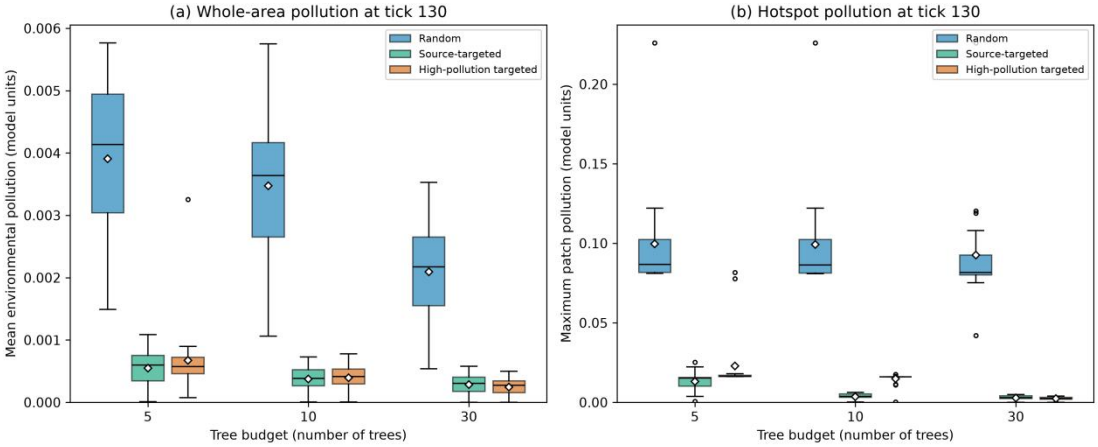



**Figure 4. Mean environmental pollution (a) and maximum patch pollution (b) at tick 130 across planting strategies and tree budgets.** Each boxplot summarises 20 runs. Boxes show the interquartile range, horizontal lines mark medians, and diamonds indicate means.

The relative performance of the targeted strategies also depended on the budget (Figure 5). Source-targeted planting produced lower mean environmental pollution at budgets of 5 and 10 trees, while high-pollution-targeted planting performed better at a budget of 30.

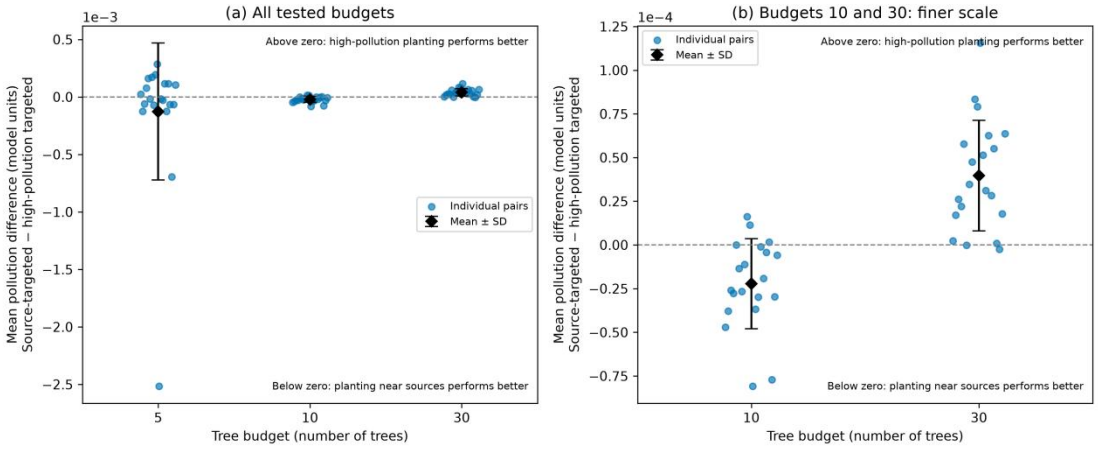



**Figure 5. Paired pollution differences between targeted strategies at tick 130.** Positive values favour high-pollution targeting. Blue dots show individual pairs; black diamonds show mean ± SD. Panel (b) uses a finer scale.

However, paired differences between the two targeted strategies included both positive and negative values at every budget, showing that neither consistently outperformed the other. Variability was particularly large at a budget of 5 trees.

Strategy comparisons should therefore consider both mean outcomes and variation across runs.

These strategy comparisons apply only to the fixed wind conditions tested; their performance under other wind strengths or directions remains untested.

## 4. Discussion and Limitations

### 4.1 Implications for Public Education

This model is intended for members of the public without specialist knowledge of modelling. It simplifies urban air pollution processes by focusing on the relationships between tree planting, pollution transport and planting location. This approach aims to make these processes easier to understand through scenario comparisons and visual results, supporting educational demonstrations with guidance from a presenter.

The experiments provide three messages for public education:

1. More trees do not necessarily produce proportional pollution reductions;

2. Lower pollution under windy conditions should not be attributed entirely to trees;

3. Planting locations matter when resources are limited.

### 4.2 Limitations of Model Realism

The estimated percentage reductions describe model responses and should not be interpreted as expected improvements in actual air quality. Trees in the model remove pollution without changing airflow. A research found increased pollution accumulation beneath tree canopies during the leaf-on period on Symonds Street in Auckland, suggesting that canopies may restrict ventilation within street canyons. Real vegetation can therefore affect both pollution removal and dispersion, while this model represents only removal[6].

The model also excludes buildings, terrain, land use, differences between vegetation types, and changing weather conditions[7]. Each tree has a fixed removal capacity, and pollution is represented as a single substance. The estimated percentage reductions therefore describe model responses and should not be interpreted as expected improvements in actual air quality.

### 4.3 Uncertainty and Future Improvements

Repeated runs captured variation caused by random source layouts and planting locations, but did not address all sources of uncertainty. Only selected planting intensities, wind transport ratios, and budgets were tested. Emission intensity, the

diffusion coefficient, and tree removal capacity remained fixed. Sensitivity findings therefore apply only to the tested ranges and model settings.

Future work could first test these fixed parameters, then introduce different vegetation removal capacities or simplified effects of buildings on airflow. Supporting actual planting decisions would also require calibration and validation using local environmental data.

Reference


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